Hein Zarni Aung

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EDUCATION

Stony Brook University **Expected Graduation: May 2027**

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Algorithms, Systems Fundamentals I & II, Programming Abstractions

INTERNSHIPS

Software Engineering Intern — Clustr **Jan 2026 – Sept 2026 (expected)**

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC **Jun 2026 – Sept 2026 (expected)**

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University **Sep 2025 – Dec 2025**

PI: Abisheka Pitumpe | Team: 2

- Built a Python/Selenium pipeline to monitor newly issued SSL/TLS certificates in real time and automatically flag domains impersonating EZ-Pass tolling services, designing keyword- and heuristic-based detection logic to triage hundreds of candidate sites into real fraud vs. noise.

PROJECTS

Diary Handwriting Analysis **Python, PyTorch, Hugging Face Transformers, OpenCV**

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline (grayscale conversion, binary thresholding, horizontal ink-projection profiling), reaching ~87% reliable page transcription on personal diary pages.
- Pivoted to DANIEL, a segmentation-free architecture that transcribes full pages directly and jointly extracts named entities, to enable topic and mood-trend statistics across diary entries over time; currently in progress.

TuneAcademy **React, NumPy, Essentia, Firestore, FastAPI, Docker**

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Set up a CI/CD pipeline using GitHub Actions to lint, test, and orchestrate automated build and deploy workflows to Cloudflare Workers.
- Backend built with Python/FastAPI in Docker, Cloud Firestore, and GCP hosting.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University **Aug 2025 – Dec 2025**

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

Backend: FastAPI, Express.js, Node.js

Frontend: React, React Native, Next.js

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, Selenium